Fnu Suya

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Professional Appointments

Tenure-track Assistant Professor 08/2024 – Present
Min H. Kao Department of EECS, University of Tennessee, Knoxville
MC2 Postdoctoral Fellow 08/2023 – 07/2024
Maryland Cybersecurity Center, University of Maryland, College Park

Education

Ph.D. in Computer Science, *University of Virginia* 08/2017 – 07/2023
Thesis: *On the Limits of Data Poisoning Attacks*
Advisor: Yuan Tian (UCLA), David Evans

M.Sc. in Computer Science, *University of Virginia* 08/2017 – 12/2022
Advisor: Yuan Tian (UCLA), David Evans

Ph.D. in Computer Science, *Arizona State University* 08/2015 – 07/2017
Advisor: Paolo Papotti (EURECOM), discontinued due to advisor relocation

B.Eng. in Electrical Engineering (Honors), *China Agricultural University* 09/2010 – 07/2014

Research Interests

Trustworthy Machine Learning, Machine Learning for Security

Internship Experience

Research Intern, *Qualcomm*, San Diego (with Aleksei Triastcyn on robust FL) 05/2021– 08/2021
Research Intern, *Amazon AWS*, New York City (with MohamadAli Torkamani on GNNs) 01/2021 – 04/2021
Research Intern, *Bosch AI Center*, Pittsburgh (with Anit Kumar Sahu on black-box attacks) 06/2020 – 08/2020
Research Intern, *Tsinghua University*, China (with Bo Bai on wireless communication) 08/2014 – 02/2015

Selected Publications

2026a S. Hoque, J. Zhang, J. Sun, **F. Suya**. "Token Inflation: How Dishonest Providers Can Overcharge for Large Language Model Usage". In: *NeurIPS 2026*.
2026b H. Rahaman, Q. Xu, Z. Haider, P. Chakraborty, S. Bhunia, **F. Suya**. "(A)iSpy: Parasitic Trojans for Machine Learning Infrastructure". In: *ACM CCS 2026*.
2026c T. Zhang, **F. Suya**, R. Jha, C. Zhang, V. Shmatikov. "Adversarial Hubness in Multi-Modal Retrieval". In: *IEEE S&P (Oakland) 2026*.
2026d S. Amgain, D. Lobo, A. Chatterjee, S. Bhunia, **F. Suya**. "HAMLOCK: HArdware-Model LOgically Combined attacK". In: *USENIX Security 2026*.
2024 F. Suya, A. Suri, T. Zhang, S. Hong, Y. Tian, D. Evans. "SoK: Pitfalls in Evaluating Black-Box Attacks". In: *IEEE SaTML 2024*.
2023 F. Suya, X. Zhang, Y. Tian, D. Evans. "What Distributions are Robust to Indiscriminate Poisoning Attacks for Linear Learners?" In: *NeurIPS 2023*.

2022 Y. Tian, **F. Suya**, F. Xu, D. Evans. "Stealthy Backdoors as Compression Artifacts". In: *IEEE TIFS 2022*.

Conference and Journal Papers

Google Scholar ID: OmLIG8EAAAAJ

2027 J. Zhang, Y. He, K. Cai, H. Zhao, **F. Suya**, Y. Tian. "RogueMerge: Robust and Unified Attacks against LLM Model Merging". In: *IEEE S&P (Oakland) 2027*.

2026a A. Al Nomaan Nafi, H. Rahaman, Z. Haider, T. Mahfuz, **F. Suya**, S. Bhunia, P. Chakraborty. "DAASH: A Meta-Attack Framework for Synthesizing Effective and Stealthy Adversarial Examples". In: *CVPR 2026*.

2026b S. Amgain, D. Lobo, A. Chatterjee, S. Bhunia, **F. Suya**. "HAMLOCK: HArdware-Model LOgically Combined attacK". In: *USENIX Security 2026*.

2026c G. Du, A. N. Khan, R. Zhou, X. Liu, D. Chakrabarti, **F. Suya**, X. Li. "Greedy Decoding Is Not Precision-Invariant: Cross-Precision Output Divergence in LLM Inference". In: *Transactions on Machine Learning Research (TMLR)*.

2026d S. Hoque, J. Zhang, J. Sun, **F. Suya**. "Token Inflation: How Dishonest Providers Can Overcharge for Large Language Model Usage". In: *NeurIPS 2026*.

2026e H. Rahaman, Q. Xu, Z. Haider, P. Chakraborty, S. Bhunia, **F. Suya**. "(A)iSpy: Parasitic Trojans for Machine Learning Infrastructure". In: *ACM CCS 2026*.

2026f T. Zhang, **F. Suya**, R. Jha, C. Zhang, V. Shmatikov. "Adversarial Hubness in Multi-Modal Retrieval". In: *IEEE S&P (Oakland) 2026*.

2024 F. Suya, A. Suri, T. Zhang, S. Hong, Y. Tian, D. Evans. "SoK: Pitfalls in Evaluating Black-Box Attacks". In: *IEEE SaTML 2024*.

2023a F. Suya, X. Zhang, Y. Tian, D. Evans. "What Distributions are Robust to Indiscriminate Poisoning Attacks for Linear Learners?" In: *NeurIPS 2023*.

2023b Y. Tian, **F. Suya**, A. Suri, F. Xu, D. Evans. "Manipulating Transfer Learning for Property Inference". In: *CVPR 2023*.

2022 Y. Tian, **F. Suya**, F. Xu, D. Evans. "Stealthy Backdoors as Compression Artifacts". In: *IEEE TIFS*.

2021 F. Suya, S. Mahloujifar, A. Suri, D. Evans, Y. Tian. "Model-Targeted Poisoning Attacks with Provable Convergence". In: *ICML 2021*.

2020a F. Suya, J. Chi, D. Evans, Y. Tian. "Hybrid Batch attacks: Finding Black-box Adversarial Examples with Limited Queries". In: *USENIX Security 2020*.

2020b J. Wang, M. Luo, **F. Suya**, J. Li, Z. Yang, Q. Zheng. "Scalable Attack on Graph Data by Injecting Vicious Nodes". In: *ECML-PKDD 2020*.

2019 Y. Chen, M. Zha, N. Zhang, D. Xu, Q. Zhao, X. Feng, K. Yuan, **F. Suya**, Y. Tian, K. Chen, X. Wang, W. Zou. "Demystifying Hidden Privacy Settings in Mobile Apps". In: *IEEE S&P (Oakland) 2019*.

2016 F. Suya, Y. Shi, B. Bai, W. Chen, J. Zhang, K. B. Letaief, S. Zhou. "Optimal Stochastic Power Control with Compressive CSI Acquisition for Cloud-RAN". In: *GlobalSIP 2016*.

Preprints

2026a J. Huang, S. Amgain, C. Gong, Y. Pang, **F. Suya**. *Demystifying Prompt Stealing Attacks against Synthetic Video*. arXiv preprint arXiv:2611.00102.

2026b B. Yan, Q. Zhang, Z. Kan, **F. Suya**, S. Wang. *PROVISO: Provenance-Informed Action Selection for Secure LLM Agents*. arXiv preprint arXiv:2610.27115.

2026c K. Cai, K. Zhang, Z. Xiang, J. Zhang, A. Alwan, **F. Suya**, Y. Tian. *Hearing the Whispers: Black-Box Membership Inference Attacks on Finetuned TTS Models*. arXiv preprint arXiv:2609.01723.

2026d J. Zhang, P. Yu, Z. Lin, K. Cai, **F. Suya**, Y. Tian. *ImageAuditor: Membership Inference Attack against Image-based Retrieval-Augmented Generation*. arXiv preprint arXiv:2606.03354.

2026e A. Al Nomaan Nafi, **F. Suya**, S. Bhunia, P. Chakraborty. *LASH: Adaptive Semantic Hybridization for Black-Box Jailbreaking of Large Language Models*. arXiv preprint arXiv:2605.21362.

2026f K. Cai, R. Song, J. Zhang, K. Zhang, P. Bodapati, A. Yu, **F. Suya**, M. Rostami, J. Ma, Y. Tian. *What-If World: A Causal Benchmark for General World Models in Embodied Scenarios*. arXiv preprint arXiv:2605.27589.

Workshop Papers and Posters

2023a A. Kinfe, C. Jung, K. Lin, M. Clyburn, **F. Suya**. "HackWrt: Network Traffic-Based Eavesdropping of Handwriting". In: *CPS-IoT Week 2023*.

2023b F. Suya, X. Zhang, Y. Tian, D. Evans. "When Can Linear Learners be Robust to Indiscriminate Poisoning Attacks?" In: *ICML AdvML-Frontiers Workshop 2023*.

2022 E. Rose, **F. Suya**, D. Evans. "Poisoning Attacks and Subpopulation Susceptibility". In: *VISxAI workshop 2022 (Best Paper Award)*.

2018 F. Suya, D. Evans, Y. Tian. "Poster: Adversaries Don't Care About Averages: Batch Attacks on Black-Box Classifiers". In: *IEEE S&P (Oakland) 2018*.

2017 F. Suya, Y. Tian, D. Evans, P. Papotti. "Query-limited Black-box Attacks to Classifiers". In: *N(eur)IPS MLSec Workshop 2017 (Spotlight Presentation)*.

Talks and Presentations

1. An Adversarial Perspective on Machine Learning Pipeline
(Invited Talk) Warren B. Nelms Annual IoT Conference 2024, Gainesville, FL 12/2024
2. What Distributions are Robust to Indiscriminate Poisoning Attacks for Linear Learners?
NeurIPS, New Orleans, LA 12/2023
3. When Can Linear Learners be Robust to Indiscriminate Poisoning Attacks?
ICML Workshop on AdvML-Frontiers, Online 07/2023
4. Model-Targeted Poisoning Attacks with Provable Convergence
ICML, Online 07/2021
5. Hybrid Batch Attacks: Finding Black-box Adversarial Examples with Limited Queries
Usenix Security, Online 08/2020
6. Adversaries Don't Care About Averages: Batch Attacks on Black-Box Classifiers
IEEE S&P (Oakland), San Francisco, CA 05/2018
7. Query-limited Black-box Attacks to Classifiers
N(eur)IPS workshop on MLSec, Long Beach, CA 12/2017
8. Optimal Stochastic Power Control with Compressive CSI Acquisition for Cloud-RAN
GlobalSIP, Washington DC 12/2016

Mentoring Experience

- 2026 - Present, Ony Hoque, PhD student, University of Tennessee, Knoxville (co-advised with Jinyuan Stella Sun)
- 2025 - Present, Sanskar Amgain, PhD student, University of Tennessee, Knoxville
- 2025 - Present, Qipan Xu, MS student, University of Tennessee, Knoxville
- 2024 - 2026, Gomathi Lakshmanan, MS student, University of Tennessee, Knoxville (graduated)

Teaching Experience

- Machine learning (UTK COSC 522), Instructor F2026
- Introduction to Cybersecurity (UTK COSC 366), Instructor S2026
- Machine learning (UTK COSC 522), Instructor F2025
- Adversarial Machine learning (UTK COSC 594), Instructor S2025
- Introduction to Cybersecurity (UTK COSC 366), Instructor F2024
- Learning Theory (UVA CS 6501-005), TA S2019
- Cryptography (UVA CS 6501-009), TA S2019
- Game Theory (ASU CSE 556), TA F2016
- Introduction to C++ Programming (ASU CSE 100), TA F2015 – S2016

- Introduction to Programming Languages (ASU CSE 240), TA F2015 – S2016

Grants

- Energy & Gateway Internal Award, ORIED, University of Tennessee (Role: PI; Amount: $60,000) 2026
- Gift from Arista Networks in support of the TRUST-AI Seminar Series (Amount: $10,000) 2026
- Support for Affiliated Research Teams (StART), UT-Oak Ridge Innovation Institute (Role: PI; Amount: $200,000; Funding Rate: 6%) 2025 – 2026
- AI TechX Award, AI Tennessee Initiative, University of Tennessee, jointly with Cisco (Role: PI; Amount: $60,000) 2025
- ORIED Internal Award in support of the TRUST-AI Seminar Series, ORIED, University of Tennessee (Amount: $10,000) 2025

Honors & Awards

- MC2 Postdoctoral Fellowship: awarded annually to one individual by the Maryland Cybersecurity Center (MC2) 2023
- Outstanding/Top Reviewer: ICLR 2022, NeurIPS 2022, 2023
- CS Graduate Research Award, University of Virginia 2018
- CS Department Fellowship, University of Virginia 2017
- NSF Travel Grant 2016
- CIDSE Doctoral Fellowship, Arizona State University 2015
- Outstanding Student Scholarship, China Agricultural University 2011 – 2013

Service

Associate Editor	Journal of Hardware and Systems Security (Springer), 2026 – Present
PC/Reviewer	IEEE S&P, Usenix Security, CCS, NDSS, IEEE EuroS&P, AsiaCCS, ICML, NeurIPS, ICLR, ICCV, CVPR, ECCV, AISTATS, IJCAI, AAAI
Journal Reviewer	Artificial Intelligence, TMLR

Skills

Programing	Python, PyTorch, TensorFlow, Matlab, C, C++
Languages	Mongolian (native), Mandarin, English

References

- Available upon request